

Анализ трафика в Wireshark

Лабораторная работа №3

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Цели и задачи работы

Изучение кадров Ethernet и анализ PDU протоколов канального, транспортного и прикладного уровней стека TCP/IP с использованием анализатора Wireshark.

Выполнение работы

Получение параметров сетевого соединения

Адаптер беспроводной локальной сети Беспроводная сеть:

```
DNS-суффикс подключения . . . . . :  
Описание. . . . . : Intel(R) Wi-Fi 6E AX211 160MHz  
Физический адрес. . . . . : F8-FE-5E-6F-7B-EC  
DHCP включен. . . . . : Да  
Автонастройка включена. . . . . : Да  
Локальный IPv6-адрес канала . . . : fe80::da3b:4057:9ef4:1e28%7(Основной)  
IPv4-адрес. . . . . : 192.168.1.69(Основной)  
Маска подсети . . . . . : 255.255.255.0  
Аренда получена. . . . . : 13 декабря 2025 г. 12:39:21  
Срок аренды истекает. . . . . : 14 декабря 2025 г. 12:39:18  
Основной шлюз. . . . . : 192.168.1.1  
DHCP-сервер. . . . . : 192.168.1.1  
IAID DHCPv6 . . . . . : 133758558  
DUID клиента DHCPv6 . . . . . : 00-01-00-01-30-47-94-1A-10-FF-E0-21-D6-B4  
DNS-серверы. . . . . : 192.168.1.1  
NetBios через TCP/IP. . . . . : Включен  
PS C:\Users\Victor\Documents> |
```

Рис. 1: Результат выполнения команды ipconfig

```
PS C:\Users\Victor\Documents> ping 192.168.1.1

Обмен пакетами с 192.168.1.1 по 32 байтами данных:
Ответ от 192.168.1.1: число байт=32 время=2мс TTL=64
Ответ от 192.168.1.1: число байт=32 время=2мс TTL=64
Ответ от 192.168.1.1: число байт=32 время=1мс TTL=64
Ответ от 192.168.1.1: число байт=32 время=3мс TTL=64

Статистика Ping для 192.168.1.1:
    Пакетов: отправлено = 4, получено = 4, потеряно = 0
    (0% потерь)
Приблизительное время приема-передачи в мс:
    Минимальное = 1мсек, Максимальное = 3 мсек, Среднее = 2 мсек
PS C:\Users\Victor\Documents> |
```

Рис. 2: Проверка доступности шлюза

ICMP Echo Request / Reply

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	ASUSTekCOMPU_78:b6:...	Broadcast	ARP	42	who has 192.168.1.10? Tell 192.168.1.1
8	1.025440	ASUSTekCOMPU_78:b6:...	Broadcast	ARP	42	who has 192.168.1.10? Tell 192.168.1.1
11	2.047906	ASUSTekCOMPU_78:b6:...	Broadcast	ARP	42	who has 192.168.1.10? Tell 192.168.1.1
21	3.071926	ASUSTekCOMPU_78:b6:...	Broadcast	ARP	42	who has 192.168.1.10? Tell 192.168.1.1
24	3.262827	ASUSTekCOMPU_78:b6:...	Intel_6f:7b:ec	ARP	42	who has 192.168.1.69? Tell 192.168.1.1
25	3.262855	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:...	ARP	42	192.168.1.69 is at f8:fe:5e:6f:7b:ec
28	4.095938	ASUSTekCOMPU_78:b6:...	Broadcast	ARP	42	who has 192.168.1.10? Tell 192.168.1.1
32	5.222190	ASUSTekCOMPU_78:b6:...	Broadcast	ARP	42	who has 192.168.1.10? Tell 192.168.1.1
59	6.143759	ASUSTekCOMPU_78:b6:...	Broadcast	ARP	42	who has 192.168.1.10? Tell 192.168.1.1
62	7.167778	ASUSTekCOMPU_78:b6:...	Broadcast	ARP	42	who has 192.168.1.10? Tell 192.168.1.1
90	12.479003	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 91)
91	12.481736	192.168.1.1	192.168.1.69	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=64 (request in 90)
94	13.484590	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 95)
95	13.486852	192.168.1.1	192.168.1.69	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=64 (request in 94)
109	14.581131	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 110)
110	14.582146	192.168.1.1	192.168.1.69	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=64 (request in 109)
119	15.515669	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 120)
120	15.519033	192.168.1.1	192.168.1.69	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=64 (request in 119)
215	31.423895	ASUSTekCOMPU_78:b6:...	Intel_6f:7b:ec	ARP	42	who has 192.168.1.69? Tell 192.168.1.1
216	31.423927	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:...	ARP	42	192.168.1.69 is at f8:fe:5e:6f:7b:ec
291	45.771474	82:81:04:02:d3:54	Broadcast	ARP	42	who has 192.168.1.1? Tell 192.168.1.145
441	59.581429	ASUSTekCOMPU_78:b6:...	Intel_6f:7b:ec	ARP	42	who has 192.168.1.69? Tell 192.168.1.1
442	59.581458	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:...	ARP	42	192.168.1.69 is at f8:fe:5e:6f:7b:ec


```

> Frame 90: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF {3ABBA4AF-DF76-4939-B087-FC37529B3966}, id 0
< Ethernet II, Src: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec), Dst: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)
  Destination: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)
    ....0. .... = LG bit: Globally unique address (factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Source: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)
    ....0. .... = LG bit: Globally unique address (factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0000)
  [Stream index: 1]
  > Internet Protocol Version 4, Src: 192.168.1.69, Dst: 192.168.1.1
  > Internet Control Message Protocol
    Type: 8 (Echo (ping) request)
    Code: 0
    Checksum: 0x4d5a [correct]
    [Checksum Status: Good]
    Identifier (BE): 1 (0x0001)
    Identifier (LE): 256 (0x0100)
    Sequence Number (BE): 1 (0x0001)
    Sequence Number (LE): 256 (0x0100)
    [Sequence request: 31]
  > Data (32 bytes)
  
```

Рис. 3: ICMP Echo Request

ICMP Echo Request / Reply

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	ASUSTekCOMPU_78:b6:...	Broadcast	ARP	42	Who has 192.168.1.10? Tell 192.168.1.1
8	1.025449	ASUSTekCOMPU_78:b6:...	Broadcast	ARP	42	Who has 192.168.1.10? Tell 192.168.1.1
11	2.047906	ASUSTekCOMPU_78:b6:...	Broadcast	ARP	42	Who has 192.168.1.10? Tell 192.168.1.1
21	3.071926	ASUSTekCOMPU_78:b6:...	Broadcast	ARP	42	Who has 192.168.1.10? Tell 192.168.1.1
24	3.262827	ASUSTekCOMPU_78:b6:...	Intel_6f:7b:ec	ARP	42	Who has 192.168.1.69? Tell 192.168.1.1
25	3.262855	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:...	ARP	42	192.168.1.69 is at f8:fe:5e:6f:7b:ec
28	4.095838	ASUSTekCOMPU_78:b6:...	Broadcast	ARP	42	Who has 192.168.1.10? Tell 192.168.1.1
32	5.222190	ASUSTekCOMPU_78:b6:...	Broadcast	ARP	42	Who has 192.168.1.10? Tell 192.168.1.1
59	6.143759	ASUSTekCOMPU_78:b6:...	Broadcast	ARP	42	Who has 192.168.1.10? Tell 192.168.1.1
62	7.167778	ASUSTekCOMPU_78:b6:...	Broadcast	ARP	42	Who has 192.168.1.10? Tell 192.168.1.1
→ 90	12.479003	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 91)
91	12.481736	192.168.1.1	192.168.1.69	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=64 (request in 90)
94	13.484590	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 95)
95	13.488052	192.168.1.1	192.168.1.69	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=64 (request in 94)
109	14.501131	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 110)
110	14.502146	192.168.1.1	192.168.1.69	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=64 (request in 109)
119	15.515669	192.168.1.69	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 120)
120	15.519033	192.168.1.1	192.168.1.69	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=64 (request in 119)
215	31.423895	ASUSTekCOMPU_78:b6:...	Intel_6f:7b:ec	ARP	42	Who has 192.168.1.69? Tell 192.168.1.1
216	31.423927	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:...	ARP	42	192.168.1.69 is at f8:fe:5e:6f:7b:ec
201	45.771474	82:81:04:02:d9:54	Broadcast	ARP	42	Who has 192.168.1.1? Tell 192.168.1.145
441	59.581429	ASUSTekCOMPU_78:b6:...	Intel_6f:7b:ec	ARP	42	Who has 192.168.1.69? Tell 192.168.1.1
442	59.581458	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:...	ARP	42	192.168.1.69 is at f8:fe:5e:6f:7b:ec
639	87.740225	ASUSTekCOMPU_78:b6:...	Intel_6f:7b:ec	ARP	42	Who has 192.168.1.69? Tell 192.168.1.1
640	87.740237	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:...	ARP	42	192.168.1.69 is at f8:fe:5e:6f:7b:ec

Frame 91: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF{3ABB4A4F-DF76-4939-8087-FC3752983966}, id 0 Ethernet II, Src: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2), Dst: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec) - Destination: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec) 0... = 10 bit: Globally unique address (factory default) 0... = 10 bit: Individual address (unicast) - Source: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2) 0... = 10 bit: Globally unique address (factory default) 0... = 10 bit: Individual address (unicast) Type: IPv4 (0x0800) [Stream index: 1] Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.69 Internet Control Message Protocol - Type: 0 (Echo (ping) reply) - Code: 0 - Checksum: 0x555a [correct] [Checksum Status: Good] - Identifier (BE): 1 (0x0001) - Identifier (LE): 256 (0x0100) - Sequence Number (BE): 1 (0x0001) - Sequence Number (LE): 256 (0x0100) [Request frame: no] [Response time: 2,733 ms] Data (32 bytes)	0000 f8 fe 5e 6f 7b 0010 00 3c f1 ec 00 0020 01 45 00 00 55 0030 67 68 69 6a 6b 0040 77 61 62 63 64
--	--

Ethernet (eth) 14.6 Gbps Пакець 721 - Отобразити: 25 (3.5%) Профіль: Default

Рис. 4: ICMP Echo Reply

120	15.519055	192.168.1.1	192.168.1.69	ILMP	/4 echo (ping) reply id=0x0001, seq=4/1024, ttl=64 (request in 119)		
215	31.423895	ASUSTekCOMPU_78:b6:f2	Intel_6f:7b:ec	ARP	42 Who has 192.168.1.69? Tell 192.168.1.1		
216	31.423927	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:f2	ARP	42 192.168.1.69 is at f8:fe:5e:6f:7b:ec		
291	45.771474	82:81:04:02:d3:54	Broadcast	ARP	42 Who has 192.168.1.1? Tell 192.168.1.145		
441	59.581429	ASUSTekCOMPU_78:b6:f2	Intel_6f:7b:ec	ARP	42 Who has 192.168.1.69? Tell 192.168.1.1		
442	59.581458	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:f2	ARP	42 192.168.1.69 is at f8:fe:5e:6f:7b:ec		
639	87.740225	ASUSTekCOMPU_78:b6:f2	Intel_6f:7b:ec	ARP	42 Who has 192.168.1.69? Tell 192.168.1.1		
640	87.740237	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:f2	ARP	42 192.168.1.69 is at f8:fe:5e:6f:7b:ec		
753	105.776239	82:81:04:02:d3:54	Broadcast	ARP	42 Who has 192.168.1.1? Tell 192.168.1.145		
809	115.899467	ASUSTekCOMPU_78:b6:f2	Intel_6f:7b:ec	ARP	42 Who has 192.168.1.69? Tell 192.168.1.1		
810	115.899490	Intel_6f:7b:ec	ASUSTekCOMPU_78:b6:f2	ARP	42 192.168.1.69 is at f8:fe:5e:6f:7b:ec		
874	124.719775	a2:1f:8e:51:22:03	Broadcast	ARP	42 Who has 192.168.1.1? Tell 192.168.1.24		
> Frame 215: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF{3ABB4A4F-DF76-4939-8087-FC37529B3966}, id 0							0000 f8 fe 5e 6f 7b ec
> Ethernet II, Src: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2), Dst: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)							0010 08 00 06 04 00 00 00
* Destination: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec)							0020 00 00 00 00 00 00 00
....00 = LG bit: Globally unique address (factory default)							
....00 = IG bit: Individual address (unicast)							
* Source: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)							
....00 = LG bit: Globally unique address (factory default)							
....00 = IG bit: Individual address (unicast)							
Type: ARP (0x0806)							
[Stream Index: 1]							
> Address Resolution Protocol (request)							
Hardware type: Ethernet (1)							
Protocol type: IPv4 (0x0800)							
Hardware size: 6							
Protocol size: 4							
Opcode: request (1)							
Sender MAC address: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)							
Sender IP address: 192.168.1.1							
Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00)							
Target IP address: 192.168.1.69							

Рис. 5: ARP-запрос

```
PS C:\Users\Victor\Documents>
PS C:\Users\Victor\Documents> ping ya.ru

Обмен пакетами с ya.ru [77.88.44.242] с 32 байтами данных:
Ответ от 77.88.44.242: число байт=32 время=19мс TTL=248
Ответ от 77.88.44.242: число байт=32 время=18мс TTL=248
Ответ от 77.88.44.242: число байт=32 время=21мс TTL=248
Ответ от 77.88.44.242: число байт=32 время=18мс TTL=248

Статистика Ping для 77.88.44.242:
    Пакетов: отправлено = 4, получено = 4, потеряно = 0
    (0% потерь)

Приблизительное время приема-передачи в мс:
    Минимальное = 18мсек, Максимальное = 21 мсек, Среднее = 19 мсек
PS C:\Users\Victor\Documents> |
```

Рис. 6: Ping доменного имени

227	17.584410	192.168.1.69	188.184.67.127	HTTP	502 GET / HTTP/1.1
229	17.641196	188.184.67.127	192.168.1.69	HTTP	932 HTTP/1.1 200 OK (text/html)
233	17.657844	192.168.1.69	188.184.67.127	HTTP	443 GET /favicon.ico HTTP/1.1
237	17.789564	188.184.67.127	192.168.1.69	HTTP	248 HTTP/1.1 200 OK (image/vnd.microsoft.icon)
274	20.890715	192.168.1.69	188.184.67.127	HTTP	562 GET /hypertext/www/TheProject.html HTTP/1.1
280	20.945081	188.184.67.127	192.168.1.69	HTTP	1044 HTTP/1.1 200 OK (text/html)
320	26.552122	192.168.1.69	188.184.67.127	HTTP	583 GET /hypertext/README.html HTTP/1.1
330	26.603615	188.184.67.127	192.168.1.69	HTTP	1485 HTTP/1.1 200 OK (text/html)

.....

Frame 227: 502 bytes on wire (4016 bits), 502 bytes captured (4016 bits) on interface \Device\NPF {3AB84A4F-DF76-4939-8087-FC3752983966}, i

Ethernet II, Src: Intel 6f:7b:ec (f8:fe:5e:6f:7b:ec), Dst: ASUSTekCOMPU 78:b6:f2 (c8:7f:54:78:b6:f2)

Internet Protocol Version 4, Src: 192.168.1.69, Dst: 188.184.67.127

Transmission Control Protocol, Src Port: 49295, Dst Port: 80, Seq: 1, Ack: 1, Len: 448

Source Port: 49295

Destination Port: 80

[Stream index: 4]

[Stream Packet Number: 4]

[Conversation completeness: Complete, WITH_DATA (31)]

[TCP Segment Len: 448]

Sequence Number: 1 (relative sequence number)

Sequence Number (raw): 1229692954

[Next Sequence Number: 449 (relative sequence number)]

Acknowledgment Number: 1 (relative ack number)

Acknowledgment number (raw): 2145874767

0101 = Header Length: 20 bytes (5)

Flags: 0x018 (PSH, ACK)

Window: 513

[Calculated window size: 131328]

[Window size scaling factor: 256]

Checksum: 0xc3ff [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

[Timestamps]

[SEQ/ACK analysis]

TCP payload (448 bytes)

Hypertext Transfer Protocol

Рис. 7: HTTP-трафик

174	-68.852560	192.168.1.69	192.168.1.1	DNS	72 Standard query 0x4201 A login.vk.com	
175	-68.852560	192.168.1.69	192.168.1.1	DNS	72 Standard query 0x158f HTTPS login.vk.com	
176	-68.837484	192.168.1.1	192.168.1.69	DNS	104 Standard query response 0x4261 A login.vk.com A 93.186.237.1 A 95.213.	
177	-68.837013	192.168.1.1	192.168.1.69	DNS	128 Standard query response 0x158f HTTPS login.vk.com SOA ns1.vkontakte.ru	
246	-62.906209	192.168.1.69	192.168.1.1	DNS	75 Standard query 0xcb28 A play.google.com	
247	-62.906036	192.168.1.69	192.168.1.1	DNS	75 Standard query 0x8a77 HTTPS play.google.com	
249	-62.892818	192.168.1.1	192.168.1.69	DNS	171 Standard query response 0xcb28 A play.google.com A 209.85.233.139 A 20	
250	-62.883364	192.168.1.1	192.168.1.69	DNS	125 Standard query response 0x8a77 HTTPS play.google.com SOA ns1.google.co	
265	-61.647580	192.168.1.69	192.168.1.1	DNS	77 Standard query 0x3529 HTTPS line-mode.cern.ch	
266	-61.647150	192.168.1.69	192.168.1.1	DNS	77 Standard query 0x4201 A login.vk.com	
▶ Frame 175: 72 bytes on wire (576 bits), 72 bytes captured (576 bits) on interface \Device\NPF{3AB84A4F-DF76-4939-8087-FC37529B3966}, id 0						0000
▶ Ethernet II, Src: Intel 6f:7b:ec (f8:fe:5e:6f:7b:ec), Dst: ASUSTEKCOMPU 78:b6:f2 (c8:7f:54:78:b6:f2)						0010
▶ Internet Protocol Version 4, Src: 192.168.1.69, Dst: 192.168.1.1						0020
▶ User Datagram Protocol, Src Port: 61899, Dst Port: 53						0030
Source Port: 61899						0040
Destination Port: 53						
Length: 38						
Checksum: 0x83ce [unverified]						
[Checksum Status: Unverified]						
[Stream Index: 16]						
[Stream Packet Number: 1]						
▶ [Timestamps]						
UDP payload (30 bytes)						
▶ Domain Name System (query)						

Рис. 8: DNS-запрос

651	79.038739	192.168.1.69	173.194.221.95	QUIC	1292 Initial, DCID=0afdf89a7722bb4f, PKN: 1, CRYPTO, PING, PING, PING
652	79.038764	192.168.1.69	173.194.221.95	QUIC	1292 Initial, DCID=0afdf89a7722bb4f, PKN: 2, CRYPTO, CRYPTO, CRYPTO
653	79.038836	192.168.1.69	173.194.221.95	QUIC	122 0-RTT, DCID=0afdf89a7722bb4f
657	79.060407	173.194.221.95	192.168.1.69	QUIC	82 Initial, SCID=eafdf89a7722bb4f, PKN: 1, ACK
658	79.060691	173.194.221.95	192.168.1.69	QUIC	1292 Initial, SCID=eafdf89a7722bb4f, PKN: 2, ACK, PADDING
659	79.061430	173.194.221.95	192.168.1.69	QUIC	1292 Initial, SCID=eafdf89a7722bb4f, PKN: 3, CRYPTO, PADDING
660	79.061430	173.194.221.95	192.168.1.69	QUIC	341 Protected Payload (KP0)
661	79.061430	173.194.221.95	192.168.1.69	QUIC	993 Protected Payload (KP0)

Frame 652: 1292 bytes on wire (10336 bits), 1292 bytes captured (10336 bits) on interface \Device\NPF_{3AB84A4F-DF76-4939-8087-FC37529B396E}

Ethernet II, Src: Intel_E6:7b:ec (f8:fe:5e:6f:7b:ec), Dst: ASUSTekCOMPU_78:b6:f2 (c8:7f:54:78:b6:f2)

Internet Protocol Version 4, Src: 192.168.1.69, Dst: 173.194.221.95

User Datagram Protocol, Src Port: 53350, Dst Port: 443

Source Port: 53350
 Destination Port: 443
 Length: 1258
 Checksum: 0x520b [unverified]
 [Checksum Status: Unverified]
 [Stream index: 30]
 [Stream Packet Number: 2]
 [Timestamps]
 UDP payload (1250 bytes)

QUIC IETF

QUIC Connection Information
 [Packet Length: 1250]
 1... = Header Form: Long Header (1)
 .1... = Fixed Bit: True
 ..00 = Packet Type: Initial (0)
 [.... 00.. = Reserved: 0]
 [.... ..00 = Packet Number Length: 1 bytes (0)]
 Version: 1 (0x00000001)
 Destination Connection ID Length: 8
 Destination Connection ID: 0afdf89a7722bb4f
 Source Connection ID Length: 0
 Token Length: 70
 Token: 0065b348b4552f5ee4851e2275116268cff2cfd7af8da1d6a17690e271bcd0affb9d91b80b8d77dff4ca92d8933c0eedc4b48b23dfa3f83c75f10290b32b0357
 Length: 1161
 [Packet Number: 2]
 Payload [..]: 7062aca8eae0db296dd3f41fd050a77c6596d3c86f187e7d86d0790970a68588802114cd8d9841977f0525a6afed4e36303941dec74e0e462c18b9d982

CRYPTO
 CRYPTO
 CRYPTO
 CRYPTO
 PING
 PING
 PING
 CRYPTO
 CRYPTO
 CRYPTO

Установка TCP-соединения

200	17.895346	192.168.1.69	188.184.67.127	TCP	60 63245 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM WS=2
201	17.895346	192.168.1.69	188.184.67.127	HTTP	587 GET /hypertext/WWW/People.html HTTP/1.1
209	17.947169	188.184.67.127	192.168.1.69	TCP	66 80 → 63245 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=2
210	17.947197	192.168.1.69	188.184.67.127	TCP	54 63245 → 80 [ACK] Seq=1 Ack=1 Win=131328 Len=0
211	17.955341	188.184.67.127	192.168.1.69	TCP	54 80 → 63814 [ACK] Seq=1 Ack=534 Win=64128 Len=0
212	17.956972	188.184.67.127	192.168.1.69	TCP	1514 80 → 63814 [ACK] Seq=1 Ack=534 Win=64128 Len=1460 [TCP PDU reassembled in 2
213	17.956972	188.184.67.127	192.168.1.69	TCP	1514 80 → 63814 [PSH, ACK] Seq=1461 Ack=534 Win=64128 Len=1460 [TCP PDU reassembled
214	17.956972	188.184.67.127	192.168.1.69	TCP	1514 80 → 63814 [ACK] Seq=2921 Ack=534 Win=64128 Len=1460 [TCP PDU reassembled
215	17.956972	188.184.67.127	192.168.1.69	TCP	1514 80 → 63814 [PSH, ACK] Seq=4381 Ack=534 Win=64128 Len=1460 [TCP PDU reassembled
216	17.956989	192.168.1.69	188.184.67.127	TCP	54 63814 → 80 [ACK] Seq=534 Ack=5841 Win=131328 Len=0
217	17.957128	188.184.67.127	192.168.1.69	TCP	1514 80 → 63814 [ACK] Seq=5841 Ack=534 Win=64128 Len=1460 [TCP PDU reassembled
218	17.957128	188.184.67.127	192.168.1.69	HTTP	376 HTTP/1.1 200 OK (text/html)
219	17.957128	188.184.67.127	192.168.1.69	TCP	54 80 → 63814 [FIN, ACK] Seq=7623 Ack=534 Win=64128 Len=0
220	17.957137	192.168.1.69	188.184.67.127	TCP	54 63814 → 80 [ACK] Seq=534 Ack=7624 Win=131328 Len=0
221	17.957343	192.168.1.69	188.184.67.127	TCP	54 63814 → 80 [FIN, ACK] Seq=534 Ack=7624 Win=131328 Len=0
226	18.014374	188.184.67.127	192.168.1.69	TCP	54 80 → 63814 [ACK] Seq=7624 Ack=535 Win=64128 Len=0
228	18.078454	192.168.1.69	173.194.220.188	TCP	55 49623 → 5228 [ACK] Seq=1 Ack=1 Win=512 Len=1
230	18.094532	173.194.220.188	192.168.1.69	TCP	66 5228 → 49623 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2
231	18.202474	192.168.1.69	173.194.220.188	TCP	55 49622 → 5228 [ACK] Seq=1 Ack=1 Win=512 Len=1
233	18.223242	173.194.220.188	192.168.1.69	TCP	66 5228 → 49622 [ACK] Seq=1 Ack=2 Win=1046 Len=0 SLE=1 SRE=2
248	18.559439	192.168.1.69	4.207.247.139	TCP	66 [TCP Retransmission] 63817 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=2
249	19.629920	192.168.1.69	149.154.167.99	TLSv1.2	187 Application Data
251	19.676373	149.154.167.99	192.168.1.69	TLSv1.2	167 Application Data
253	19.731553	192.168.1.69	149.154.167.99	TCP	54 10560 → 443 [ACK] Seq=1615 Ack=1121 Win=2053 Len=0
- Frame 213: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface \Device\NPF_{3ABB4A4F-DF76-4939-80B7-FC37529B396E} - Ethernet II, Src: ASUSTekCOMPU_78:b6:f2 (c8:f7:54:78:b6:f2), Dst: Intel_6f:7b:ec (f8:fe:5e:6f:7b:ec) - Internet Protocol Version 4, Src: 188.184.67.127, Dst: 192.168.1.69 - Transmission Control Protocol, Src Port: 80, Dst Port: 63814, Seq: 1461, Ack: 534, Len: 1460 - Source Port: 80 - Destination Port: 63814 [Stream index: 5] [Stream Packet Number: 7] [Conversation completeness: Complete, WITH_DATA (31)] [TCP Segment Len: 1460] - Sequence Number: 1461 (relative sequence number) - Sequence Number (raw): 2576710090 [Next Sequence Number: 2921 (relative sequence number)] - Acknowledgment Number: 534 (relative ack number) - Acknowledgment number (raw): 1349161635 0101 ... = Header Length: 20 bytes (5) - Flags: 0x018 (PSH, ACK) - Window: 501 [Calculated window size: 64128] [Window size scaling factor: 128] - Checksum: 0xc58e [unverified] [Checksum Status: Unverified] - Urgent Pointer: 0 [Timestamps] [SEQ/ACK analysis] - TCP payload (1460 bytes) [Reassembled PDU in frame: 218] - TCP segment data (1460 bytes)					0000 f8 f
					0010 05 d
					0020 01 4
					0030 01 f
					0040 57 6
					0050 64 0
					0060 73 6
					0070 73 6
					0080 6d 7
					0090 20 6
					00a0 67 6
					00b0 73 2
					00c0 72 6
					00d0 79 7
					00e0 61 7
					00f0 20 6
					0100 6d 6
					0110 61 2
					0120 73 2
					0130 68 6
					0140 46 3
					0150 6c 3
					0160 41 3
					0170 52 4
					0180 72 6
					0190 32 2
					01a0 64 5
					01b0 70 2

Рис. 10: TCP handshake

График потока TCP

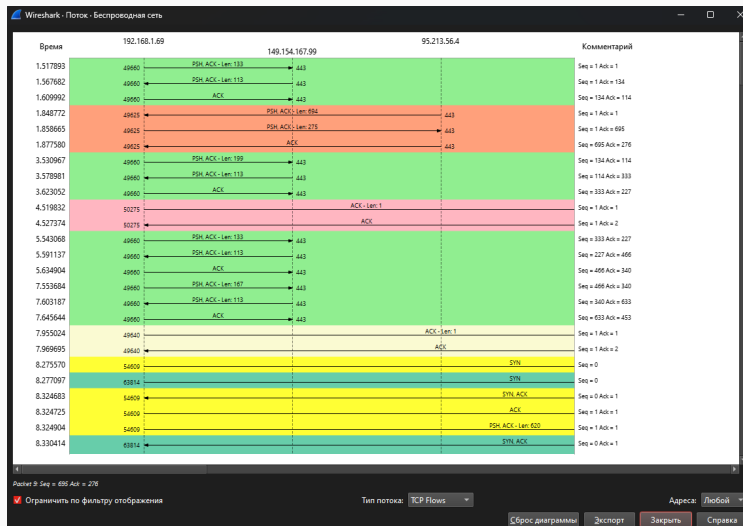


Рис. 11: График потока TCP

Вывод

- Изучены принципы MAC- и IP-адресации
- Выполнен захват и анализ сетевого трафика
- Исследованы протоколы:
 - ARP, ICMP
 - TCP, UDP, QUIC
 - HTTP и DNS
- Практически подтверждена работа TCP handshake и механизмов надёжной передачи данных